CHAPTER I

The Institution

ACLC College of Mandaue is located in the North Reclamation Area, Mandaue City, Cebu. Established as a learning institution, it has grown to become a prominent educational provider in the region. Over the years, ACLC College of Mandaue has expanded its academic offerings and facilities to cater to a diverse student population. With a commitment to fostering holistic development, the institution has continuously evolved to meet the demands of modern education. Due to its reputation for quality education and accessible programs, it has become a preferred choice for students seeking higher learning. The college's amenities include:modern computer laboratories, a comprehensive library, air-conditioned classrooms, a spacious multi-purpose hall, student lounge areas, administrative offices, and a canteen. The college is currently exploring The Development of a Clinic Information and Medical Inventory Management System for ACLC College of Mandaue, examining how the integration of digital health services and information impacts their well-being, academic performance, and overall college experience.

The System

The *Development of a Clinic Information and Medical Inventory Management System* is a browser-based system supported by centralized data storage. It is designed to improve the efficiency and accuracy of clinic information, particularly in managing student medical logs, medical inventory.

The system focuses on processing student medication requests and managing medical inventory within the clinic. It allows students to conveniently inquire about available medications and quickly locate the items they need. By maintaining a centralized database, the system ensures that all medications and inventory data are properly organized and easily accessible.

For administrative users, the system provides essential inventory management features. The admin can query medications availability, monitor stock levels, and track usage. The system can automatically generate notifications and reports for items that are nearly out of stock, as well as identify the most frequently used Medications.

For clinic staff, the system provides essential inventory management functionalities. It enables staff to check medication availability, monitor stock levels, and track usage accurately. Additionally, the system can automatically generate notifications and reports for items that are nearly out of stock and identify the most frequently used medications.

Purpose of the Project

Ama Computer Learning Center College of Mandaue currently relies on traditional and manual procedures in managing clinic information. As observed by the proponents, the clinic primarily uses logbook records and paper-based documentation for monitoring student visits, medical procedures, and inventory of medications. These manual processes often result in inefficiencies such as difficulty in retrieving records, time-consuming searches for information, and an increased risk of human error.

Students may experience delays during clinic transactions, particularly when

verifying the availability of medications. Since inventory monitoring is performed manually, clinic staff must physically check stock levels, which may lead to inaccuracies or late detection of shortages. In some instances, students only become

aware that a medications or medication is unavailable after making an inquiry, causing inconvenience and disruption in service.

In response to these challenges, this project aims to develop *a Clinic Information and Medical Inventory Management System* that will assist the institution in transitioning from manual to digital clinic management. The system is designed to improve the recording, storage, and retrieval of student medical logs and clinic information through a centralized database. By automating documentation processes, the system reduces clerical errors and enhances the accuracy and reliability of clinic records.

Furthermore, the system provides an efficient mechanism for managing medical inventory. It enables clinic personnel to easily monitor medications and medication availability, track stock levels, and receive automated notifications for items that are nearly out of stock. This proactive feature ensures that the clinic can address inventory shortages before they affect student services. The system also allows quick and systematic searching of medications, eliminating the need for manual checking and reducing waiting time for students.

Problem Definition

General Problem

Ama Computer Learning Center College of Mandaue continues to rely on manual and paper-based procedures in managing clinic information. The clinic primarily uses logbooks and printed records for documenting student visits, medical procedures, and monitoring medications inventory. Although these traditional methods have been

used for years, they are prone to human error, inefficient record management, and time-consuming processes.

Using a manual strategy, clinic personnel spend considerable time recording information, searching for student medical logs, and verifying medications or medication availability. These processes demand much effort, patience, and accuracy from clinic staff. Delays may occur during clinic transactions, especially when retrieving records or checking inventory manually. Furthermore, manual monitoring increases the possibility of miscalculations, overlooked stock shortages, and inconsistent documentation.

These challenges affect not only clinic personnel but also students who rely on timely and efficient medical assistance. Clinic-related concerns and issues are further explained in the specific problems below.

Specific Problems

1. **Time-consuming procedures.** Significant time is consumed in recording student visits, documenting procedures, and searching for logbook entries.
2. **Prone to human error.** Manual documentation and computations may lead to incomplete records, incorrect entries, or inaccurate data.
3. **Difficulty in retrieving records.** Tracking student medical history requires manual searching through logbooks and printed files.
4. **Inefficient inventory monitoring.** medications availability is verified through physical checking, which may cause delays and inaccuracies.
5. **Late detection of low stocks.** The absence of an automated alert system prevents early identification of nearly out-of-stock medications.

6. **Lost or misplaced records.** Paper-based files and logbooks are vulnerable to loss, damage, or misplacement.
7. **Duplication of records.** Rewriting of records may occur when staff assume files are missing or cannot be located quickly.
8. **No centralized data storage.** The clinic lacks a reliable digital database for secure, organized, and easily retrievable information.
9. **Damaged printed files/records.** Printed documents are susceptible to wear and tear over time.
10. **Delays in student service.** Students may experience inconvenience due to slow processes in verifying records and medication availability.
11. **Limited tracking of medication usage.** Manual methods restrict the clinic's ability to analyze frequently used medications.
12. **Inconsistent reporting.** Reports prepared manually may lack accuracy, consistency, and professional presentation.

Objectives of the Study

General Objective

The main objective of this conducted study is to design and develop a *Development of a Clinic Information and Medical Inventory Management System* that will improve the efficiency, accuracy, and reliability of clinic operations at Ama Computer Learning Center College of Mandaue, particularly in managing student medical records, medications inventory.

Specific Objectives

1. **To reduce the time consumed in clinic transactions.**

   The proposed system automates record-keeping and inventory monitoring, allowing clinic personnel to input and retrieve data efficiently rather than relying on manual logbooks.

2. **To minimize human errors in documentation.**

   The system ensures accurate recording of student visits, medical procedures, and inventory data through structured digital forms.

3. **To improve retrieval of student medical records.**

   Searching and tracking student clinic history will become easier using system-based search and filtering functions.

4. **To enhance inventory management.**

   The system enables clinic staff to monitor medications availability and stock levels in real time.

5. **To provide early notification of low stocks.**

   The system automatically detects and alerts administrators regarding medications that are nearly out of stock.

6. **To prevent loss or misplacement of records.**

   All clinic data, logs, and inventory records are securely stored in a centralized database.

7. **To avoid duplication of records.**

   The system validates entries and maintains organized digital records, reducing the likelihood of redundant documentation.

8. **To secure confidential clinic information.**

   Access to sensitive data is restricted through password protection and user authorization levels.

9. **To monitor medications usage.**

   The system tracks and identifies the most frequently used medications for better planning and procurement.

10. **To generate accurate and presentable reports.**

    The system produces organized reports related to student visits, procedures performed, inventory status, and stock movement.

11. **To improve clinic service efficiency.**

    Faster processing of inquiries and transactions contributes to better service delivery to students.

Relevance of the Study

The relevance of this study is to allow the institution to recognize the differences between using a manual clinic management approach and a browser-based digital system. Through this study, the clinic administration can compare the existing paper-based processes with the proposed Development of a Clinic Information and Medical Inventory Management System and evaluate the corresponding advantages, improvements, and limitations of each method.

The Administration / System Handler

For the administration or authorized system handler, this study is relevant as it introduces a more efficient and reliable method of managing clinic operations. The proposed system provides a centralized platform for recording student clinic visits, monitoring medical procedures, and managing medications inventory. It enables the administrator to easily track stock levels, receive automated alerts for low supplies, and generate accurate reports. This contributes to better decision-making, improved inventory control, and enhanced overall clinic management.

Students

For the students, the relevance of this study lies in the improvement of clinic service efficiency. With the implementation of the proposed system, students can experience faster processing of clinic-related transactions and inquiries. The system allows quick verification of medications availability, reducing waiting time and uncertainty. Additionally, accurate digital records ensure better documentation of student clinic visits and medical assistance received.

The Proponents

For the proponents, this study is relevant as it serves as a practical application of their academic knowledge and technical skills. The development of the system allows the proponents to evaluate their competencies in system analysis, design, development, and implementation. It also helps identify areas for further improvement in both technical expertise and problem-solving abilities.

Future Researchers

This study may serve as a valuable reference for future researchers who intend to conduct similar studies related to digitalization, clinic management systems, or inventory systems. The documentation can provide insights, conceptual guidance, and methodological ideas that may support the enhancement and expansion of future projects.

Scope and Limitations of the Study

Scope

The *Development of a Clinic Information and Medical Inventory Management System* is an automated browser- based system designed to support the clinic operations of Ama Computer Learning Center College of Mandaue. The system primarily focuses on managing student medical records, clinic procedures, and medications inventory.

The system provides a centralized data storage mechanism, ensuring that all data or information entered into the system is automatically saved and securely maintained within the database. This centralized approach improves record organization, accessibility, and reliability. Furthermore, the system is password-encrypted and authenticated by user roles, preventing unauthorized access and reducing the risk of confidential data leakage.

The system is designed to be used by authorized personnel, specifically the **administrator (admin/system handler)** and **students** for inquiry-related functions.

Admin / System Handler Functionalities

- Manage user access and system authentication
- Record and process student medication
- requests Approve, reject, or disburse medications based on student requests
- Add, update, and monitor medications inventory
- View and track medications availability
- Manage Staff Accounts
- Receive automated notifications for low or nearly out-of-stock items
- Generate and view reports related to:
  – Student visit logs
  – Clinic procedures
  – Inventory status
  – medications usage
- Search and retrieve student medical records efficiently

Clinic Staff Operational Functionalities

- Record student clinic visits and basic medical information
- Assist in documenting clinic procedures and services provided
- Check and verify medication availability
- Dispense medications and record usage
- Monitor inventory levels for reference purposes
- Receive notifications for low-stock medications

Clinic Operations Covered by the System

The system covers processes related to student medication request management and inventory monitoring of medications.. It supports digital recording, searching, tracking, and reporting functionalities necessary for efficient clinic management.

Limitations of the Study

System Limitations

Online medical consultations are not included in the system.
The system is limited to clinic record management and medications inventory monitoring only.

The system does not perform medical diagnosis or treatment recommendations.
Integration with external pharmacy or hospital systems is not supported.
The system relies on accurate manual input by authorized personnel.

User Limitations

The system is accessible only to authorized personnel — the Admin and Clinic Staff. Students do not have direct access to the system. Clinic staff handles all student-related transactions, including recording medication requests and documenting clinic visits on behalf of students.

Students are not authorized to modify clinic records or inventory data.
Only the admin/system and clinic staff handler is permitted to add, edit, update, or delete records and inventory information

CHAPTER II

REVIEW OF RELATED LITERATURE AND STUDIES

The transition from traditional, paper-based workflows to the adopting of digital clinic information systems represents a fundamental shift in how educational institutions manage health records and resources.

This chapter discusses the related literature and studies that support the shift towards a **Clinic Information and Medical Inventory Management System**. It highlights the move from manual logbooks to a semi-automated, browser-based environment, focusing on the resulting improvements in operational efficiency, data security, and inventory accuracy within a clinical setting.

Definition of Key Concepts

Automation and Semi-Automation refers to the use of technology to perform tasks with minimal human intervention to reduce manual effort and improve consistency. According to **Smith (2020)**, automation systems are specifically designed to streamline repetitive processes and ensure data uniformity. While this system is technically semi-automatic because it relies on manual data entry from authorized staff, it provides automated functions such as low-stock notifications and report generation.

Adopting Digital Clinic Information Systems involves the implementation of a browser-based solution supported by centralized data storage. This digital approach is specifically designed to manage student medical logs and medical

procedures. According to **Laudon and Laudon (2020)**, the transition to digital information systems allows for a "digital firm" environment where essential business relationships and assets are managed through digital means, effectively replacing traditional logbooks with a searchable, digital history.

Medical Inventory Management is the digital monitoring of medication availability and stock levels. According to **Stair and Reynolds (2018)**, effective information systems provide the framework for accurate resource tracking and organizational control. By adopting a digital system, clinic personnel can track usage accurately and receive proactive alerts when supplies reach critical levels, ensuring that student services are not interrupted.

Importance of Adopting Digital Clinic Information Systems

The adoption of digital clinic information systems plays a vital role in improving organizational efficiency and accuracy within a school environment. According to Johnson (2021), automated systems reduce the likelihood of human error by standardizing data entry and eliminating manual recording processes. By replacing traditional logbooks with a digital framework, the institution ensures that data is accurately recorded, securely stored, and easily retrieved when needed. This directly aligns with the concept of Automation, where technology is utilized to perform tasks with minimal human intervention to ensure consistency.

In educational institutions like ACLC College of Mandaue, adopting this digital transition is essential in managing student medical records and ensuring the reliability of clinic operations. Digital systems allow for real-time monitoring of

medication availability, faster processing of student inquiries, and improved security for sensitive health data. This shift addresses the vulnerabilities of manual systems, such as misplaced records or damaged printed files, by establishing a Digital Clinic Information System—a browser-based solution supported by centralized data storage that replaces traditional physical history with a searchable digital record.

Furthermore, digital adoption enhances productivity by reducing the time required to complete administrative tasks. Brown (2019) stated that organizations that adopt automated systems experience increased operational efficiency and improved service delivery. In a clinic setting, this is evidenced by the system's ability to perform Medical Inventory Management, providing automated notifications for low-stock items and generating instant reports. This ensures proactive management and better student assistance by ensuring that medical supplies are tracked accurately and services remain uninterrupted.

Thematic Discussion

1. Automation and Accuracy in Clinic Records

Accuracy is a primary advantage of adopting digital clinic information. Manual recording methods are often prone to human errors such as incorrect data entry, incomplete information, and duplication. According to Lee (2020), automated systems minimize these errors by validating inputs and ensuring consistency in

data recording.

By transitioning to structured digital forms within the clinic's browser-based environment, the system reduces clerical errors and enhances the overall reliability of student medical logs. This ensures that every documented medical procedure is precise and consistent with the established concepts of Automation, where technology standardizes the data collection process.

2. Efficiency and Real-Time Monitoring of Inventory

Efficiency refers to the ability to complete tasks quickly and effectively through Medical Inventory Management. Automated systems significantly improve efficiency by reducing processing time and simplifying administrative workflows. While manual inventory monitoring requires staff to physically check stock levels often leading to the late detection of shortages, the proposed semi-automatic system enables clinic personnel to monitor medication availability in real time.

This proactive approach includes automated alerts for low supplies, ensuring that the clinic can address inventory needs before they affect service delivery, thereby fulfilling the operational goals of Adopting Digital Clinic Information Systems.

3. Data Management and Security through Digital Adoption

Centralized databases allow organizations to store large amounts of data securely, which is a cornerstone of the Adopting Digital Clinic Information Systems strategy. Manual systems rely on physical logbooks and paper-based

documentation, which are vulnerable to loss, damage, and misplacement.

In contrast, digital adoption provides secure digital storage, ensuring that confidential student information and medical history are protected through password encryption and user authorization levels. This centralized approach ensures that all medications and inventory data are properly organized and easily accessible, moving the institution away from the limitations of manual record-keeping toward a more resilient digital framework.

Comparison of Manual and Automated Systems

Based on the institutional needs at ACLC College of Mandaue, the following comparison highlights the differences between traditional methods and the proposed system.

Aspect	**Manual System**	**Automated System**
Accuracy	Prone to errors	High accuracy
Efficiency	Time-consuming	Fast processing
Storage	Physical (Logbooks)	Digital database
Retrieval	Difficult and slow	Easy and instant
Security	Low (Vulnerable to damage)	High with Access Control

Related Studies

- **Local Studies**

The transition toward digital health management is a growing trend in the Philippine educational landscape. Abejo et al. (2025) explored web-based solutions for clinic data management integrated with forecasting AI for tracking medical records, demonstrating how technology can predict institutional needs. Additionally, Bolaños et al. (2024) addressed specific challenges in modernizing inventory systems in primary healthcare facilities, noting that digital tools are essential for resource-constrained environments. Locally, Jamiri et al. (2022) developed a school clinic management system for Ramon Enriquez High School, highlighting the significant administrative benefits gained from moving away from manual, paper-based logbooks.

- **International Studies**

Global research further validates the effectiveness of the adopting of digital clinic information. Anderson (2020) found that computerized record systems reduced data entry errors by more than 50%, providing significantly more reliable data than traditional manual methods. Similarly, Martinez (2021) revealed that student monitoring systems improved tracking accuracy and emphasized the critical importance of real-time data access for institutional safety. The impact of integrated technology extends to the supply chain; Akindipe (2024) studied how integrated inventory systems directly improve medical supply chain accuracy. Finally, Nguyen et al. (2020) evaluated the implementation of electronic health records, identifying that the success of such systems depends on how effectively they address the specific contingency factors of the institution.

Synthesis of the Study

The reviewed literature consistently demonstrates that automation significantly improves accuracy, efficiency, and data management. While manual systems are limited by human errors and physical constraints, semi-automated systems provide reliable and organized solutions through digital processing. This study builds upon existing research by applying these digital management principles specifically to the **Clinic Information and Medical Inventory Management System for ACLC College of Mandaue**. By integrating the concepts of digital adoption and automated inventory alerts, the proposed system addresses the gaps identified in previous manual strategies.

Conclusion

In conclusion, the adopting of digital clinic information plays a crucial role in enhancing the performance of administrative systems. The transition from manual to digital clinic management is expected to reduce clerical errors, improve record retrieval, and enhance the tracking of medical supplies through structured inventory monitoring.

The reviewed literature provides strong support for the implementation of this system, establishing a firm foundation for evaluating its effectiveness in an institutional setting.

CHAPTER III

Technical Background

The contents of this chapter provide additional information for readers to be more familiar with the computer architecture and technologies used for the system and how the project works.

Current System

The current system at the clinic for medical procedures and inventory management is a manual strategy. Healthcare staff and administrators process transactions manually by writing all information — including student details, medical records, procedure logs, inventory levels, expiration dates, and usage reports — on record books, log sheets, or pieces of paper. When students request medications, staff manually check stock availability, record dispensing, and compute usage manually. Upon restocking or audits, inventory is tallied by hand, leading to errors, stockouts, expirations, and delays in student care.

Organizational Chart

An organizational chart is a diagram representation of the internal structure of the workplace of a company or organization. It shows the roles, responsibilities, and relationships between individuals inside the organization as a whole, or broken down into departments, units, or groups.

Figure 1: Organizational Chart

This figure shows the organizational chart of the Clinic's Medical and Inventory Management.

Technicality of the Project

The technical feasibility study confirms that the system aligns with current technology trends. Through networking, users access the system via a client-server connection by configuring TCP/IP addresses. Centralized data storage using PHP and MySQL ensures all student and inventory data is automatically saved and retrieved efficiently.

Networking

In the networking structure of the Clinic Information and Medical Inventory Management System, it features a client-server connection between computers and the server. Client computers (clinic staff and admin) manage student medication requests, check inventory availability from the database, and generate reports. Transactions from client computers are saved and monitored in the server's database. This structure highlights the interdependence of devices for seamless clinic operations.

Connection

Computers and servers are connected via TCP/IP configuration. Client IP addresses, DNS, and Default Gateway must be set to access the server database.

UTP Cat6 cables are used for reliable, high-speed connections.

Database

The database is central to operations, storing all student medication requests, clinic procedures, and inventory data in organized tables. MySQL handles creation, querying, and data management, serving as the system's data backbone. The system uses the following tables: users, categories, medications, medication_requests, medication_request_items, disbursements, and module_settings.

Software and Hardware Specification

Development Phase

- Laravel 9 (PHP Framework)
- PHP 8.0+
- MySQL (via XAMPP)
- VS Code
- Composer (PHP dependency manager)
- Bootstrap 5.3 (Frontend UI)
- Bootstrap Icons
- barryvdh/laravel-dompdf (PDF generation)
- Web Browser (Google Chrome)
- Laptop: Core i5, 8GB RAM
- Operating System: Windows 10/11
- Storage (Backup): USB Flash Drive / External HDD

Deployment Phase

- PHP MySQL (XAMPP)
- Laravel 9
- 1 Computer Units: Authorized users (Admin or Clinic Staff)
- UTP Cable Cat6 (length based on clinic layout)
- Operating System: Windows 10/11

Technology	Technology
Laravel 9	PHP web framework used for building the system's backend, routing, authentication, and database management.
PHP 8.0+	Server-side scripting language powering the system's logic and database interaction.
MySQL (XAMPP)	Relational database management system storing all clinic and inventory data in organized tables.
Bootstrap 5.3	Frontend CSS framework used for the system's responsive and modern user interface.
Bootstrap Icons	Icon library used throughout the system's UI for visual clarity.
barryvdh/laravel-dompdf	Laravel package used to generate downloadable PDF reports for restock and visit logs.
VS Code	Code editor used during development for writing and editing the system's source code.
Composer	PHP dependency manager used to install and manage Laravel packages.
Web Browser	Software (e.g., Chrome) for accessing the browser-based system.
Laptop/PC	Computing devices used for development and deployment.
Operating System	Software managing hardware and software resources (Windows

	10/11).
Flash Drives / HDD	Portable storage for system backups.
Computer Server	Central unit hosting the student and inventory database.
Computer Client	User terminals (clinic staff) accessing server services via network.
UTP Cable (Cat6)	High-speed Ethernet cable for network connections between server and client.

Table 1: Technologies Definition

Summary of Technologies

The system requires at least 1 computer unit installed with PHP MySQL (XAMPP). The authorized user — whether Admin or Clinic Staff — accesses the system through any web browser (e.g., Chrome) at localhost after starting the Apache and MySQL modules. This is an offline, browser-based system and does not require an internet connection. All data is stored and managed locally on the same machine.

Proposed System

How the Project Will Work

The system requires at least 1 computer unit: 1 for Admin (server in clinic office) and 1 for Clinic Staff (client at workstation). Units are networked using UTP Cat6 cables and TCP/IP configuration for database access. This is an offline,

browser-based system — no internet connection is required. Install PHP MySQL (XAMPP), start Apache and MySQL modules, and access the system via any browser (e.g., Chrome at localhost).

User Roles:

Admin: Manages medication inventory, categories, staff accounts, medication requests, and generates restock and visit log reports. Has full access to all system modules and can enable or disable module access for staff.

Clinic Staff: Records and processes student medication requests, checks medication availability, monitors stock levels, approves or disburses medications, and views reports based on assigned module access.

The system automates student medication request processing, real-time inventory monitoring with low stock and expiration alerts, usage tracking, and report generation replacing manual logs with a centralized, efficient database for improved clinic operations.

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